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San Francisco Crime Classification

by Andrés Felipe Gómez
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Part 1: Problem Description. Give the name of the competition you selected and write a few sentences describing the competition problem as you interpreted it. You want your writeup to be self-contained so your peer-reviewer does not need to go to Kaggle to study the competition description. Clarity is more important than detail. What's the overall goal? What does the data look like? How will the results be evaluated?

Example: "The task is to predict whether a given passenger survived the sinking of the Titanic based on various attributes including age, location of the passenger's cabin on the ship, family members, the fare they paid, and other information. Solutions are evaluated by comparing the percentage of correct answers on a test dataset."

Competition: San Francisco Crime Classification

The task is to predict what kind of crime was committed in San Francisco from 1/1/2003 to 5/13/2015, based on the following attributes:

- **Category** - category of the crime incident (only in train.csv). **This is the target variable to be predicted**.
- **Descript** - detailed description of the crime incident (only in train.csv)
- **Resolution** - how the crime incident was resolved (only in train.csv)

- **Dates** - timestamp of the crime incident
- **DayOfWeek** - the day of the week
- **PdDistrict** - name of the Police Department District
- **Address** - the approximate street address of the crime incident
- **X** - Longitude
- **Y** - Latitude

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Also, the training set and test set rotate every week, meaning week 1,3,5,7... belong to test set, week 2,4,6,8 belong to training set. Solutions are evaluated by Measuring the **multi-class loss** of the classification algorithm (**Sample Submission Benchmark**: 32.89184).

Is the problem description clear and comprehensible?

- 0 pts
  The submission makes no real attempt to describe the problem.
- 2 pts
  The submission makes an attempt to describe the problem, I don't really understand what is being described.
- 5 pts
  The problem description is clear enough; I understand what is going on.

Part 2: Analysis Approach. Write a few sentences describing how you approached the problem. What techniques did you use?

Example: "I split the data by gender and handled each class separately. For the females, I trivially classified all of them as "survived." For the males, I trained a random forest as a classifier. I ignored the pclass atribute that indicated the location of the passenger's cabin because I didn't think it was relevant."

I started by plotting some of the data in the training set. In geopolitics geographic position is often related to resources and location advantages, so odds where that **Coordinates** and **Address** would play a mayor role in making a successful prediction. The plotted data on **districts** however, spread the categories fairly evenly. **Dates** and **Days of Week** could have been related to specific circumstances and behavioral patterns of victims (i.e: pornography happened during certain dates), but most crimes where evenly spread on the time frame. **Days of week** did not seemed to make a difference either.

When I plotted **Coordinates** and **Addresses** I discovered several flaws in the data itself. A large subgroup of coordinates where bound to the same point **(-120,0),** yet the crimes related to them belong to different districts. Although this was a serious flaw, It could also be a significant factor in classification.

Is the approach to the problem described clearly? Do you have some idea how you might employ these techniques to solve the problem?

- 0 pts
  No real attempt was made to describe the approach to the problem.
- 2 pts
  There's a description, but I don't fully understand what it's saying.
- 5 pts
  The description is clear enough; I understand the approach.

Part 3: Initial Solution. Write a few sentences describing how you implemented your approach. What languages and libraries did you use? What challenges did you run into?

Example: "I partitioned the data by gender manually using Excel. I used Weka to build the random forest."

About the selected prediction algorithm, the amount of data was too much for the available memory, so I figured random forests could make better use of data samples than decision trees or logistic regression algorithms. I used **R** to build the **random forests**.

Addresses were a problem, since the train data has about 123000 different streets, paired on every row. My initial idea was to obtain the streets and store each one as a column on the munged data set, but handling that many columns turned out to be too much for my **8 GB of RAM**. So I ended up depending on **Coordinates, Dates, Districts** and **Days of Week** (Addresses could not be used as a single column cause R random forests can handle up to 52 categories per column).

As a starting point, I used an **arbitrary 2% stratified sample with replacement**. I made the sample stratified to get at least one row for every category.

Is the initial solution implementation clearly described? Are the tools, languages, and libraries used reasonable?

- 0 pts
  No significant attempt was made to answer the question.
- 2 pts
  The implementation is described, but I don't fully understand how it works.
- 5 pts
  The implementation is well-described; I understand how it works.

Part 4: Initial Solution Analysis. Write a few sentences assessing your approach. Did it work? What do you think the problems were?

Example: "My approach did not work so well, achieving a score of 0.65. This is less than the sample solution. I suspect I should not have ignored the pclass attribute."

Having an arbitrary sample that small, the results were better than expected (Frankly, I thought my first attempt would not get close to the benchmark). The first attempt had **27.07401 multi-class loss** (**Sample Submission Benchmark**: 32.89184) and was ranked **716 of 797**. Having beaten the benchmark the performance was good but not amazing. I figured a bigger sample would reduce the multi-class loss so I set to test how much my single PC would take without crashing. The result was between 20% and 40%. Having to test it, I went for a **20% sample (not stratified)** with replacement and re-did the test.

Is the initial solution analysis comprehensible and reasonable?

- 0 pts
  No significant attempt was made to assess the approach.
- 2 pts
  There's an assessment, but I don't really understand what is being described.
- 5 pts
  The assessment is adequately described; I understand what's going on.

Part 5: Revised Solution and Analysis. Write a few sentences describing how you improved on your solution, and whether or not it worked.

Example: "I included the pclass attribute and ignored the ticket number attribute. My score improved to 0.68."

Using a **20% sample (not stratified)** the performance of the algorithm showed some improvement. The submission had **26.14690 multi-class loss** and was ranked **697 of 797** (went up 19 positions).

Results suggest that a big-data implementation (i.e. map-reduce) that can handle the whole registry, could drastically improve the performance of the algorithm. Also, I'm still convinced that mapping Addresses as columns could make a significant reduction of the multi-class loss. A big-data implementation could have this under consideration as well.

Is the revised solution and analysis presented clearly?

- 0 pts
  No significant attempt was made to describe an improved solution.

- 2 pts
  There's a description, but I don't fully understand it.
- 5 pts
  The improvement is adequately described; I understand it.

Overall evaluation-Given the description provided, do you feel you would be able to reproduce the result?

- 0 pts
  No, the description is too sketchy, incomplete, or unclear for me to be able to reproduce this result.
- 2 pts
  No, but primarily because I don't have a strong enough background in the area; there is a fair amount of jargon used.
- 4 pts
  Yes, I think I could reproduce the result with some effort.
- 5 pts
  Yes, the solution is clearly and thoroughly described -- I could follow them easily.

Evaluating the analyses of others' is a great way to pick up new insights and viewpoints. What is one helpful thing you got out of reviewing this submission? (e.g. maybe ideas on analyzing solutions, incentive to go try this competition, etc.)?

(This question will not affect the submitter's score.) You're not expected to actually try and reproduce the submitter's result, but you may choose to do so. Did you try to reproduce the result?

- No
- Yes, but my result was different
- Yes, and my result was the same.

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